# Multiplying and Dividing in Scientific Notation

*Teacher's Edition*

## The Period

**MA.8.NSO.1.5** Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

**Must** Multiply and divide numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

**Must not** No within-2 limit applies here — that clarification is about addition and subtraction only, and the contrast with 4.01 is the point of the day. Significant digits are 4.04.

**Target** I can multiply and divide two numbers written in scientific notation, and write the answer in scientific notation.

**MTR** **MTR.5.1** — Notes I · **MTR.3.1** — Boards 1, 3 and 7 · **MTR.4.1** — Board 6

| Segment | Slides | Time |
|---|---|---|
| Settle in; post the learning target | 1 | 1 min |
| Warm-Up | 2–4 | 6 min |
| Notes | 5–7 | 11 min |
| Example 1 | 8–9 | 3 min |
| Your Turn | 10–11 | 3 min |
| Example 2 | 12–13 | 3 min |
| Your Turn | 14–15 | 3 min |
| Whiteboards | 16–33 | 12 min |
| Independent Set | 34 | 6 min |
| IXL | 35 | 5 min |
| **Total** | | **53 min** |

### Say these three things out loud today

1. The powers of ten never have to match when you multiply or divide. They can be as far apart as they like.
2. Multiply the fronts and add the backs; divide the fronts and subtract the backs, top minus bottom.
3. A product of two coefficients under 10 is very often over 10. Look at the coefficient before you put the board up.

## Slide by Slide

1. **Multiplying and Dividing in Scientific Notation.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q1 and Q2 are the two halves of today, side by side. *1 retrieves 4.01 — same power: add the coefficients and keep the power (yesterday). 2 retrieves 3.02 — quotient of powers with base 10: subtract the exponents (last week). 3 retrieves Unit 2 — perfect squares (2.01). Roots return in 4.05 (last unit). 4 retrieves MA.6.NSO.2.1 — dividing decimals; today every quotient starts with one of these (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. $6.2 \times 10^6$ 2. $10^5$ 3. 13 4. 3
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy both lines.** Multiply the fronts, add the backs. The powers of ten never have to match when you multiply.
6. **Notes.** Top minus bottom, always in that order. And subtracting a negative goes up.
7. **Notes.** Coefficient over 10, point left and exponent up. Coefficient under 1, point right and exponent down. Same repair as yesterday.
8. **Example 1 — Multiplying, and the answer needs finishing.** SPOKEN ONLY.
9. **Example 1 — Worked.** Regroup, multiply, add the exponents, then repair.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same shape, the other sign on the small exponent.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Multiplication and division in one expression.** SPOKEN ONLY.
13. **Example 2 — Worked.** The intermediate value is $60 \times 10^6$, which is exactly the unfinished form — and it does not have to be repaired yet, because the division is still to come.
14. **Your Turn — Same steps, your numbers. Boards up when done.** A negative exponent inside the product and a division that pushes the answer below 1.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **–17. Board 1.** $(3\times10^5)(3\times10^6)$ → ***9 × 10¹¹*** *$9 \times 10^{30}$ — the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$]; $6 \times 10^{11}$ — the coefficients were added [8.NSO.1.5: the operation is multiplication]*
18. **–19. Board 2.** A printer makes $4 \times 10^3$ copies per hour. It runs for $2.5 \times 10^2$ hours. Paper costs 3 cents a sheet. How many copies does it make? → ***1 × 10⁶*** *The paper cost of 3 cents a sheet is not needed. anything built from 3 cents — the unneeded figure was used [the question asks for a count, not a cost]; $10 \times 10^5$ — right value, coefficient not less than 10 [B1G-M 8.NSO.1.5 misconception]*
20. **–21. Board 3.** $\frac{9.6\times10^8}{1.2\times10^3}$ → ***8 × 10⁵*** *$8 \times 10^{11}$ — the exponents were added [8.NSO.1.5: division subtracts the exponents]; $8.4 \times 10^5$ — the coefficients were subtracted instead of divided [8.NSO.1.5]*

**22–23. Board 4.** $\frac{3.28\times10^{7}}{4\times10^{10}}$ → ***8.2 × 10⁻⁴*** *0.82 × 10⁻³ — right value, but 0.82 is not at least 1 [8.NSO.1.5: the coefficient satisfies 1 ≤ a < 10]; 8.2 × 10⁻³ — the point moved but the exponent did not*

**24–25. Board 5.** (4 × 10⁷)(2 × 10⁹). Which is the product? → ***C — 8 × 10¹⁶*** *(A) the exponents were subtracted, 7 − 9 [8.NSO.1.5: multiplication adds the exponents]; (B) the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: (2 × 10⁴)(3 × 10⁵) = 6 × 10²⁰]; (D) the coefficients were added instead of multiplied [8.NSO.1.5: the operation is multiplication] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**26–27. Board 6.** (5 × 10⁶)(4 × 10³). Which is the product, written in scientific notation? → ***D — 2 × 10¹⁰*** *(A) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]; (B) equal in value, but 20 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12 × 10⁹ instead of 1.2 × 10¹⁰]; (C) the exponents were multiplied and the coefficient was left over 10 [8.NSO.1.5] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

**28–29. Board 7.** $\frac{7\times10^{9}}{7\times10^{-6}}$ → ***1 × 10¹⁵*** *1 × 10³ — the negative sign on the bottom exponent was dropped [8.NSO.1.5: subtracting a negative adds]; 1 × 10⁻¹⁵ — bottom minus top [8.NSO.1.5: top minus bottom]*

**30–31. Board 8.** (8.4 × 10⁷) ÷ (n × 10²) = 4.2 × 10⁵. What is the value of n? → ***n = 2*** *4.2 — the coefficients were subtracted, 8.4 − 4.2 [8.NSO.1.5: the coefficients divide]; 0.5 — 4.2 ÷ 8.4 instead of 8.4 ÷ 4.2*

**32–33. Board 9.** For (7.2 × 10¹¹) ÷ (2.4 × 10³), Marcus writes 4.8 × 10⁸ and Sara writes 3 × 10¹⁴. Name what each one did wrong, and give the correct quotient. → ***3 × 10⁸*** *'they both got it wrong' — neither error is named [the item asks what each one did]; 4.8 × 10⁸ — Marcus's error repeated*

**34. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**35. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

## Misconceptions to Watch

**Board 1.** 9 × 10³⁰ — the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: (2 × 10⁴)(3 × 10⁵) = 6 × 10²⁰]

**Board 2.** anything built from 3 cents — the unneeded figure was used [the question asks for a count, not a cost]

**Board 3.** 8 × 10¹¹ — the exponents were added [8.NSO.1.5: division subtracts the exponents]

**Board 4.** 0.82 × 10⁻³ — right value, but 0.82 is not at least 1 [8.NSO.1.5: the coefficient satisfies 1 ≤ a < 10]

**Board 5.** the exponents were subtracted, 7 − 9 [8.NSO.1.5: multiplication adds the exponents]

**Board 6.** the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]

**Board 7.** 1 × 10³ — the negative sign on the bottom exponent was dropped [8.NSO.1.5: subtracting a negative adds]

**Board 8.** 4.2 — the coefficients were subtracted, 8.4 − 4.2 [8.NSO.1.5: the coefficients divide]

**Board 9.** 'they both got it wrong' — neither error is named [the item asks what each one did]

**Materials** Whiteboards and markers. Calculators are allowed on everything; a calculator is the fastest way to settle whether $8 \times 10^{16}$ and $8 \times 10^{63}$ could possibly both be the answer to whiteboard 5.